

When the Planner's Misestimation Inflates Your Cloud Bill

Finding the Costly Queries `pg_stat_statements` Can't See

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From XXXXXXXX X. XXXXXXXX <XXXXXXXXXXXXXXXX@XXXXXX.COM>

Subject **Segmentation fault caused by Postgres - reateplan.c:6178 - prepare_sort_from_pathkeys**

2023-12-31 03:23:12.048 PST [17763] LOG: server process (PID 19014) was terminated by **signal 11: Segmentation fault**

2023-12-31 03:23:12.048 PST [17763] DETAIL: Failed process was running:

with recursive

pks_fks as (

-- pk + fk referencing col

select

contype::text as contype,

conname,

array_length(conkey, 1) as ncol,

conrelid as resorigtbl,

col as resorigcol,

ord

from pg_constraint

left join lateral unnest(conkey) with ordinality as _(col, ord)

where contype IN ('p', 'f')



private chat with an AQO user (sensitive information hidden)

12 Jan 2024

Be careful using AQO – just remember, AI guesses, and these predictions can sometimes be totally wrong. The best results I achieved were when I used it manually, correcting only specific problematic queries. 1:41 ✓✓

We have turned on AQO across **5 000 servers** for all queries now 3:06

We love it. 3:06

Do you have some positive effect? 3:07 ✓✓

Yes. 3:15

Much faster across the board. 3:15

I.e. run the query 50% time without AQO then 50% w/ AQO then compare 3:18



The query plan is a great source of savings on your cloud bill.

It requires no extra hardware or watts — just a little thought and careful tuning.



Is it easy to spot an optimisation problem?

One production ERP database ...

Example query

```
SELECT T1.Fld100343RRef,
       (CAST(CAST((T16.Fld100345 * T1.Fld100346) AS NUMERIC(28,8)) /
              (T16.Fld100346 * T1.Fld100345) AS NUMERIC(28,8)))
FROM (SELECT T15.Fld100344RRRef AS Fld100344RRRef,
             T15.Fld100343RRRef AS Fld100343RRRef, T15.Fld100346 AS Fld100346,
             T15.Fld100345 AS Fld100345
FROM (SELECT T3.Fld100343RRRef AS Fld100343RRRef,
             T3.Fld100344RRRef AS Fld100344RRRef, MAX(T3._Period) AS MAXPERIOD,
             FROM _InfoRgi100342 T3
WHERE ((T3.Fld2488 = CAST(0 AS NUMERIC))) AND (T3._Period <=
'2025-12-04 12:49:02':timestamp AND (EXISTS(SELECT 1
FROM pg_temp.tt3319 T4
LEFT OUTER JOIN _Reference101 T5
ON (T4._Q_001_F_007RRRef = T5._IDRRRef) AND (T5.Fld2488 = CAST(0 AS NUMERIC)))
LEFT OUTER JOIN _Reference395 T6
ON (T4._Q_001_F_006RRRef = T6._IDRRRef) AND (T6.Fld2488 = CAST(0 AS NUMERIC)))
LEFT OUTER JOIN _Reference366X1 T7
ON (T4._Q_001_F_006RRRef = T7._IDRRRef) AND (T7.Fld2488 = CAST(0 AS NUMERIC)))
WHERE ((1=1 OR 1=0) AND (T4._Q_001_F_001RRRef =
'216e030f\332\2618\333\021\356\222\212\342\004z\226'::bytea) AND
(T5.Fld53293RRRef = '214\2150\271\351\244\310\000B<\227\003{-\356e'::bytea)
AND ((T7.Fld59861RRRef < '201Fnu\257Q\301Fp\0270\214\214\001{'::bytea)
OR (T7.Fld59861RRRef = '201Fnu\257Q\301Fp\0270\214\214\001{'::bytea)))
AND ((T3.Fld100343RRRef = T5.Fld53293RRRef) AND
(T3.Fld100344RRRef = T6.Fld101340RRRef)))
UNION ALL SELECT 1
FROM pg_temp.tt3319 T8
LEFT OUTER JOIN _Reference101 T9
ON (T8._Q_001_F_007RRRef = T9._IDRRRef) AND (T9.Fld2488 = CAST(0 AS NUMERIC)))
LEFT OUTER JOIN _Reference101 T10
ON (T9.Fld53293RRRef = T10._IDRRRef) AND (T10.Fld2488 = CAST(0 AS NUMERIC)))
LEFT OUTER JOIN _Reference395 T11
ON (T8._Q_001_F_006RRRef = T11._IDRRRef) AND (T11.Fld2488 = CAST(0 AS NUMERIC)))
LEFT OUTER JOIN _Reference366X1 T12
ON (T8._Q_001_F_006RRRef = T12._IDRRRef) AND (T12.Fld2488 = CAST(0 AS NUMERIC)))
WHERE ((1=1 OR 1=0) AND (T8._Q_001_F_001RRRef =
'216e030f\332\2618\333\021\356\222\212\342\004z\226'::bytea) AND
(T9.Fld53293RRRef = '214\2150\271\351\244\310\000B<\227\003{-\356e'::bytea)
```

Statistics

- 49500 distinct queryids
- 560 queries longer than 1s.
- query text: 33 – 126 000 chars
- execution time: 1 ms – 26 s
- calls: 1 – 5 075
- blocks read: 0 – 3.5 M (≈27 GB)
- plan nodes: 1 – 195
- never executed nodes: 0 – 52



Closing the blind spot in query statistics

`pg_stat_statements`

<code>calls</code>	<code>[10 000 000]</code>
<code>mean_exec_time</code>	<code>[100.0]</code>
<code>total_exec_time</code>	<code>[1 000 000 000]</code>
<code>rows</code>	<code>[10 000 000]</code>
<code>shared_blks_read</code>	<code>[50 000 000]</code>

278 hours of execution time,
near the top of any ranking by
total time.

What we need:

- Plan quality
- Filter efficiency
- Join method choice
- Missing indexes
- work_mem impact
- ...



Assessing the instance's potential

- 1** Estimation error (rows estimated / rows actual)
is it worth manipulating table statistics?
- 2** Selectivity of scans and joins (rows removed by filter)
do we have enough indexes?
- 3** SubPlans' share of the execution time
is it time to rewrite the queries?
- 4** Spilling of intermediate results
should queries get more memory?
- 5** Plan stability
would a generic plan be a better idea?



Postgres extension pg_track_optimizer

executor
finishes a query

→
estimated vs actual
rows, per node

→
running stats
in shared memory

→
one view:
pg_track_optimizer

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metrics per query [ID]

0 restarts

to start tracking: shared
memory via the DSM
registry, no preload needed

QueryID

Modified Query ID to
reduce the number of
different entries

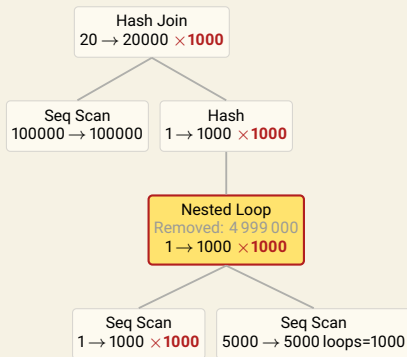
Persistent

flushed to disk on backend
exit, loaded back after a
restart



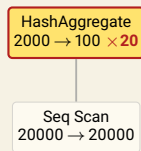
Nuance 1: Queries are different

Two joins over a filtered scan



6 nodes · 2 joins · 226 ms

GROUP BY over a scan



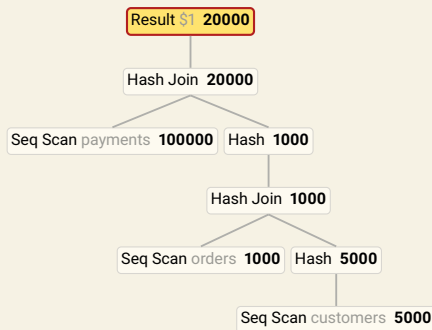
2 nodes · 0 joins · 1.4 ms



Nuance 2: Parameter might be in undefined state

```
PREPARE q(int, bool) AS  
  SELECT o.id, p.id FROM orders o  
    JOIN customers c ON c.id = o.cust JOIN payments p ON p.cust = c.id  
  WHERE $1;
```

EXECUTE q(true)



EXECUTE q(false)



Nuance 3: statistics' stochastic nature

```
ANALYZE t; -- 40 times, the data never change
EXPLAIN SELECT * FROM t WHERE x = 77; -- actual: 10000 rows every time
```

77 made it into the MCV list

$\approx 10\,000$ rows

Bitmap Heap Scan

21 runs of 40

77 missed the MCV list

≈ 60 rows, **off by $\times 170$**

Index Scan

19 runs of 40



Two design decisions in pg_track_optimizer

Dimensionless metrics:

per plan node, units cancel out:

- $e = \left| \ln \frac{\text{actual rows}}{\text{estimated rows}} \right|$
- $r = \frac{\text{node time}}{\text{total time}}$
- $f = \frac{\text{filtered rows}}{\text{actual rows}}$

weighted averaging over the executed nodes: by time, cost, none.

Custom RStats type:

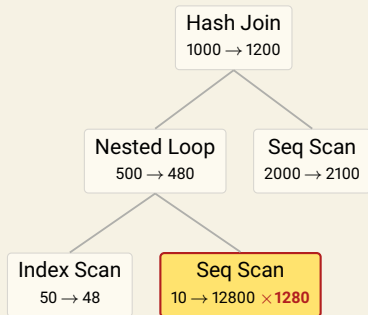
a distribution, not a number:

(count, mean, min, max, stddev)

- Updated in place on every execution (Welford): the plan is unstable, not just wrong.
- The same code path for each parameter.
- Aggregate on statistic column.



Cardinality estimation error



per-node error:

$$e = \left| \ln \frac{\text{actual}}{\text{estimated}} \right|$$

How to average across plan nodes?



How does it work in practice?

Test bed: **the Join Order Benchmark**

Leis et al., *How Good Are Query Optimizers, Really?*, PVLDB 2015

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queries over real IMDb data

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tables joined in the deepest query

Scans

many of them, including parameterised ones

Room

plenty of it, for the planner to go wrong



How the estimation error criteria work

TOP-10 by average error V/S TOP-10 by time-weighted error

```
SELECT avg.* FROM
  (SELECT queryid, query, avg_avg, twa_avg FROM pg_track_optimizer
   ORDER BY avg_avg DESC LIMIT 10) AS avg
JOIN
  (SELECT queryid FROM pg_track_optimizer
   ORDER BY twa_avg DESC LIMIT 10) AS twa
USING (queryid);
```

queryid	query	avg_avg	twa_avg
-6836119621424365027	/* 28a.sql */	3.91	2.16
3631768377448547020	/* 29c.sql */	3.77	2.51
-8544699698054799639	/* 22d.sql */	3.48	2.09
4682104415514032107	/* 7c.sql */	3.48	2.18

(4 rows)



A look inside 28a.sql

12.	0.000	0.000	↓ 15,866.0		1	→ Nested Loop (cost=673.574,16..713,608.92 rows=1 width=65) (actual rows=15,866.00 loops=1)
13.	0.000	0.000	↓ 2,641,497.5		1	→ Parallel Hash Join (cost=673.573,73..713,604.75 rows=2 width=69) (actual rows=5,282,995.00 loops=1) Hash Cond: (mk.movie_id = t.id)
14.	0.000	0.000	↓ 4.0		1	→ Parallel Seq Scan on movie_keyword mk (cost=0.00..35,789.82 rows=1,130,982 width=8) (actual rows=4,523,930.00 loops=1)
15.	0.000	0.000	↓ 47,805.0		1	→ Parallel Hash (cost=673.573,72..673,573.72 rows=1 width=61) (actual rows=47,805.00 loops=1) Buckets: 65,536 (originally 1024) Batches: 1 (originally 1) Memory Usage: 5,368kB
16.	0.000	0.000	↓ 47,805.0		1	→ Nested Loop (cost=611,718.06..673,573.72 rows=1 width=61) (actual rows=47,805.00 loops=1) Join Filter: (t.id = mk.movie_id)
17.	0.000	0.000	↓ 18,042.0		1	→ Nested Loop (cost=611,717.63..673,566.96 rows=3 width=40) (actual rows=54,126.00 loops=1)
18.	0.000	0.000	↓ 7,874.1		1	→ Parallel Hash Join (cost=611,717.20..673,551.17 rows=7 width=28) (actual rows=55,119.00 loops=1) Hash Cond: (pi.person_id = ci.person_id)
19.	0.000	0.000	↓ 94.7		1	→ Parallel Hash Join (cost=1.84..61,811.22 rows=6,553 width=4) (actual rows=620,526.00 loops=1) Hash Cond: (pi.info_type_id = it3.id)
20.	0.000	0.000	↓ 4.0		1	→ Parallel Seq Scan on person_info pi (cost=0.00..59,792.74 rows=740,474 width=8) (actual rows=2,963,664.00 loops=1)
21.	0.000	0.000	↑ 1.0		1	→ Parallel Hash (cost=1.83..1.83 rows=1 width=4) (actual rows=1.00 loops=1) Buckets: 1,024 Batches: 1 Memory Usage: 40kB
22.	0.000	0.000	↑ 1.0	- 112	1	→ Parallel Seq Scan on info_type it3 (cost=0.00..1.83 rows=1 width=4) (actual rows=1.00 loops=1) Filter: ((info)::text = 'trivia'::text) Rows Removed by Filter: 112

```

-> Parallel Hash Join (rows=6553) (actual rows=620526)
    Hash Cond: (pi.info_type_id = it3.id)
-> Parallel Seq Scan on person_info pi (rows=740474) (actual rows=2963664)
-> Parallel Hash (rows=1) (actual rows=1.00)
    -> Parallel Seq Scan on info_type it3 (rows=1) (actual rows=1)
        Filter: ((info)::text = 'trivia'::text)
        Rows Removed by Filter: 112

```

⇒ The x4 scan error grows throughout the join tree.



Filter factors: scans and joins

Scans

$$f_{scan} = \max_{scans} \left(\frac{\text{removed by filter}}{\text{actual rows}} \times \frac{\text{node time}}{\text{total time}} \right)$$

Joins

$$f_{join} = \max_{joins} \left(\frac{\text{removed by join filter}}{\text{actual rows}} \times \frac{\text{node time}}{\text{total time}} \right)$$

Hash Join (actual time=234.671 rows=289 loops=1)

Hash Cond: (o.customer_id = c.id)

Join Filter: (o.amount > c.credit_limit)

Rows Removed by Join Filter: 47542

-> Seq Scan on orders o (actual time=98.341 rows=47831 loops=1)

Filter: (status = 'completed')

Rows Removed by Filter: 952169

-> Hash (actual time=42.110 rows=100000 loops=1)

-> Seq Scan on customers c (actual time=21.543 rows=100000)



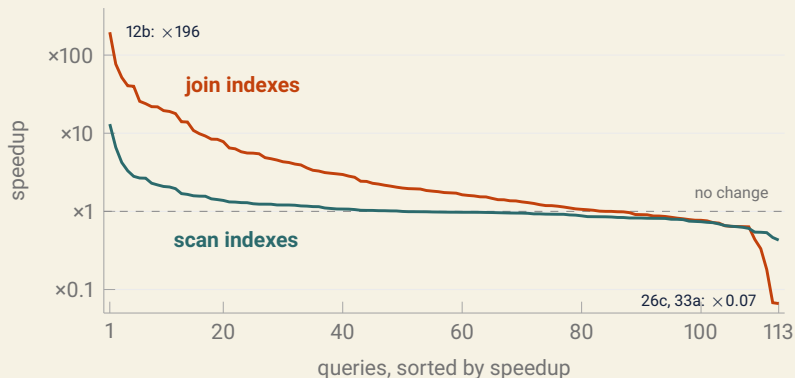
queryid	query	avg_avg	lf_avg
7987931480126446697	/* 15b.sql */	1.60	1459.35
-4244378222084090458	/* 23a.sql */	1.75	1424.58
-6679854959453081533	/* 15a.sql */	1.95	1406.36
-1139306376856509831	/* 23b.sql */	1.48	1403.23
-4904731422305343909	/* 15c.sql */	2.69	1342.81

[illegible]

[illegible]

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Effect of the indexes found



JOB, 113 queries: plain time / time with the new indexes, best to worst

